BMC Public Health

RESEARCH

Open Access

# Leveraging XGBoost and explainable AI for accurate prediction of type 2 diabetes

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**Abstract**

**Introduction** Type 2 diabetes mellitus (T2DM) poses a major public health challenge, particularly in regions with limited representation in predictive modeling studies. This research aimed to develop and interpret robust machine learning (ML) models for early T2DM risk prediction using data from the Dena Cohort in Iran.

**Methods** Data from 3,203 adults aged 35–70 years were preprocessed through outlier removal, median/mode imputation, feature selection via LightGBM, and class balancing with the Synthetic Minority Over-sampling Technique (SMOTE). Two gradient-boosting algorithms, XGBoost and CatBoost, underwent hyperparameter tuning and 10-fold cross-validation. Model performance was assessed using accuracy, F1-score, and the area under the receiver operating characteristic curve (AUC). Shapley Additive Explanations (SHAP) provided global and case-specific interpretability of predictive features.

**Results** XGBoost achieved the highest performance (accuracy = 96.07%, AUC = 99.29%), outperforming CatBoost and demonstrating substantial improvement with SMOTE balancing. Key predictors included fasting blood sugar, fatty liver, urolithiasis, red blood cell indices, and lifestyle factors such as energy drink consumption and prolonged television viewing. SHAP visualizations enhanced model transparency and facilitated individualized risk interpretation.

**Conclusion** This study demonstrates that combining advanced gradient-boosting models with SHAP explainability yields highly accurate, interpretable T2DM risk prediction in an underrepresented population. These findings support integrating interpretable ML into clinical workflows for personalized prevention and early intervention strategies.

**Keywords** Type 2 diabetes, Machine learning, XGBoost, CatBoost, Explainable AI, SHAP, Dena cohort

## Introduction

### Epidemiological burden and diagnostic challenges of type 2 diabetes

Type 2 diabetes (D.M. II) is a chronic metabolic disorder that represents one of the most significant public health concerns worldwide. Characterized by insulin resistance and progressive beta-cell dysfunction, D.M. II is responsible for considerable morbidity and mortality due to its association with various long-term complications. These complications include cardiovascular diseases, kidney failure, neuropathy, and retinopathy, leading to diminished quality of life and increasing healthcare costs. According to global estimates, the number of people living with diabetes continues to rise rapidly, with projections indicating that over 640 million individuals will be affected by 2040 [1–3].

In most cases, D.M. II begins gradually and with no or mild symptoms. This characteristic makes many patients unaware of their disease for a long time and delays

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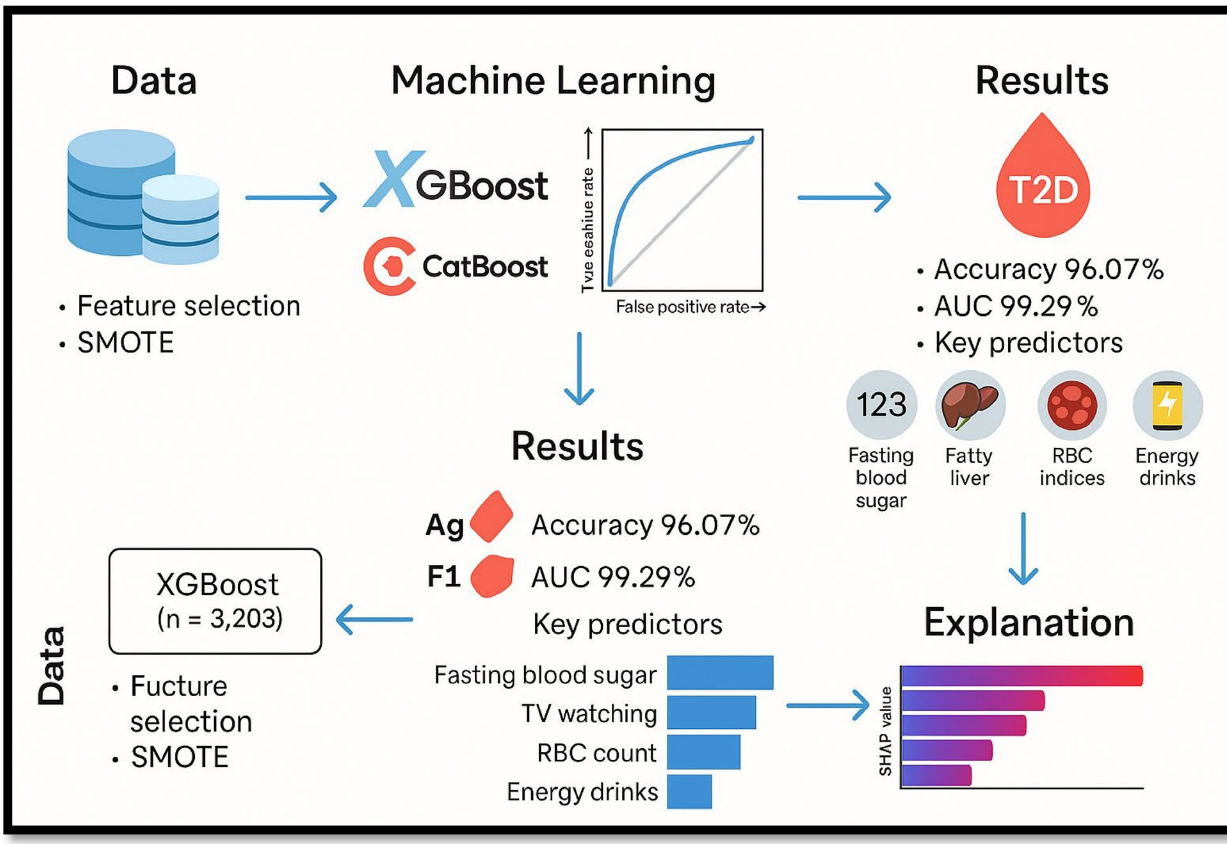
**Graphical abstract**




diagnosis. Symptoms such as thirst, frequent urination, fatigue and blurred vision may be mild or ignored, making them difficult to diagnose quickly.

Another point is the similarity of the symptoms of diabetes with other diseases; for example, fatigue, weight loss or blurred vision may be confused with the symptoms of other diseases, which further complicates accurate diagnosis. In addition, many patients enter an asymptomatic phase at the beginning of the disease, during which blood sugar levels slowly increase but the person himself does not feel any symptoms. This silent phase can last for years and can only be detected by screening tests.

Another challenge is the lack of public awareness; Many people are not informed about the risk factors, the importance of screening tests, and the early symptoms of D.M. II, and therefore seek medical attention later. In addition, in some areas, limited access to laboratory facilities and regular screening programs makes early diagnosis difficult.

In addition to delaying treatment, late diagnosis of diabetes can cause significant financial costs and health burden for the individual and society, as complications of the disease appear at an advanced stage and require more complex and expensive treatments [1, 2, 4, 5].

## Research gap and solutions

### *Challenges in prediction D.M. II*

One of the key challenges in the management of type 2 diabetes is to identify individuals at risk before the disease progresses to advanced stages. This is of great importance, as the first step in diabetes prevention and control is to identify high-risk individuals early so that preventive measures and effective interventions can be initiated and serious complications or rapid disease progression can be prevented.

Traditional risk assessment tools, such as standardized questionnaires and clinical examinations, although useful in some cases, are limited and often inadequate. These tools usually do not capture the complex and multifaceted interactions of genetic, environmental, behavioral, and lifestyle variables that influence diabetes risk. For example, genetic factors, physical activity level, diet, psychological stress, socio-economic conditions, and environmental factors are usually considered independently and separately in these tools, while in fact these factors interact and interfere with each other and have a combined and complex effect on the risk of the disease.

This limitation of traditional tools strongly emphasizes the need for new and more advanced data-driven approaches, including Big Data analysis technologies, intelligent models, and machine learning. New technologies are able to better and more accurately predict the risk of diabetes in high-risk individuals by analyzing a large amount of information and identifying hidden patterns. These approaches can focus on the complex interactions between genetics, environment, and lifestyle, and therefore provide more accurate predictions, which is much more important than traditional tools.

As a result, the use of advanced data-driven approaches and new technologies will not only improve the risk assessment process, but also provide a platform for more personalized and targeted interventions. This will increase the capacity to implement targeted prevention programs and reduce the incidence of type 2 diabetes in the community, and ultimately, will make a significant contribution to reducing the financial and health burden caused by this disease [6–8].

#### *Solution: artificial intelligence in healthcare and D.M. II prediction*

In recent years, machine learning (ML) and artificial intelligence (AI) have emerged as promising tools in different aspects of human life, especially health care, particularly for predictive analytics [9–12]. ML algorithms excel at processing large datasets, identifying complex patterns, and making accurate predictions that may not be apparent through traditional statistical methods. These capabilities make ML especially valuable for conditions like D.M II, where multiple variables interact to influence disease onset [13–15].

Several ML algorithms, including supervised learning models like decision trees, random forests, and gradient-boosting machines, have been applied to predict D.M. II in various populations. These models are trained on datasets containing input variables (e.g., demographic information, clinical measurements) and the corresponding outcomes (e.g., D.M. II diagnosis). Once trained, these models can predict whether new individuals are at risk of developing D.M. II, enabling proactive identification and intervention [15, 16].

### The need for explainable AI (XAI)

Machine learning models in healthcare have achieved high levels of predictive accuracy and play a critical role in improving diagnosis, treatment, and predicting clinical outcomes. By analyzing vast amounts of medical data, including radiology images, electronic health records, and laboratory data, these models help identify patterns and signs of disease that may be difficult or impossible for humans to detect [7, 17]. However, the inherent complexity of these models has led them to often be referred to as “black boxes,” meaning that the decision-making process and how the predictions are generated are not understandable or transparent to healthcare professionals [18, 19]. In clinical settings, trust and transparency are of paramount importance. Healthcare providers need to be able to understand the reasoning and logic behind the predictions of machine learning models to ensure that the recommendations provided are consistent with their clinical judgment and appropriate for the specific patient situation [13, 20]. This is especially important in sensitive medical decisions that can affect patients’ lives.

In this regard, explainable AI (XAI) plays a vital role. XAI refers to a set of methods and techniques that aim to increase the transparency, understandability, and trust in artificial intelligence models. These technologies allow clinicians and experts to see and analyze the reasoning behind the predictions and decisions of machine learning models, which increases trust in these systems and improves their acceptance in clinical settings [6].

Thus, while machine learning models have high predictive power, without explainability, their application in healthcare will be limited. Combining predictive accuracy with transparency and explainability is key to the success of AI in medicine, helping healthcare providers make better decisions for patients and avoiding the risks of errors from unintelligible models. XAI refers to techniques designed to make AI models more transparent and interpretable. Using XAI, the reasoning behind an ML model’s predictions can be clearer to clinicians and patients alike. These fosters trust in the model and allows healthcare providers to validate its predictions against their expertise. In the context of D.M. II, XAI can highlight the factors contributing to a patient’s predicted risk, such as elevated blood glucose levels and lifestyle factors, making it invaluable for informed decision-making [7, 13, 21, 22].

### Review of existing studies on ML and AI in D.M II prediction

Several studies have explored the application of artificial intelligence (AI) and machine learning (ML) in the prediction and management of Type 2 diabetes (D.M II). Nomura et al. (2021) conducted a comprehensive review of the role of AI in diabetes management, highlighting that AI has significant potential to enhance diabetes management through predictive analytics [21]. Similarly, Channa et al. (2021) assessed the clinical applications of AI for diabetic retinopathy, demonstrating the efficacy of AI systems in improving screening accuracy [18]. Sheng et al. (2022) provided insights into AI’s role in ocular diseases related to diabetes, indicating that AI can effectively assist in predicting and managing ocular complications. Gunasekeran et al. (2020) further explored AI’s capabilities in predicting diabetic retinopathy outcomes and found that AI tools significantly improve the prediction of diabetic retinopathy, facilitating timely intervention [1]. Dankwa-Mullan et al. (2019) investigated how AI can transform diabetes care delivery through a systematic review of AI applications. They concluded that AI technologies can improve access, accuracy, and efficiency in diabetes care [2]. Contreras and Vehi (2018) reviewed AI methods for decision support in diabetes care, emphasizing that AI can enhance clinical decision-support systems, ultimately improving diabetes management outcomes. Bhat et al. (2023) developed a framework for predicting diabetes risk using ML techniques, finding that their comprehensive approach effectively predicts diabetes risk cc. Ahmed et al. (2022) investigated the effectiveness of fused ML techniques in predicting diabetes and noted improved prediction accuracy by integrating diverse machine-learning approaches [20]. Khanam and Foo (2021) compared various ML algorithms for diabetes prediction and highlighted the effectiveness of ensemble methods [21]. Finally, Sisodia et al. (2018) evaluated the effectiveness of classification algorithms for diabetes prediction, concluding that certain algorithms provide high accuracy in predicting D.M. II [17].

More recent works have further advanced this field by integrating deep learning, optimization, and explainable AI (XAI) techniques. Several studies have demonstrated the value of combining ML with interpretable approaches, including a transparent ML framework for glioma prognosis [23], an XAI-driven predictive model for assessing hematopoietic stem cell transplantation outcomes in pediatric patients [24]. In addition, recent publications have applied cutting-edge ML and optimization methods in healthcare prediction, enhancing model accuracy and interpretability [25, 26]. These works underscore the growing importance of interpretable and high-performance models in clinical prediction tasks and support the application of SHAP in our study to improve transparency and trust in diabetes risk assessment.

### Novel contributions

This study introduces several methodological and clinical contributions that distinguish it from prior work on diabetes prediction:

- Population-specific modeling: Leveraging Dena Cohort data, representing a Middle Eastern population often underrepresented in machine learning research for diabetes risk prediction.
- Advanced algorithms: Employing XGBoost and CatBoost, two high-performance gradient-boosting algorithms, optimized with hyperparameter tuning and validated with cross-validation.
- Data balancing strategy: Applying SMOTE to address class imbalance, ensuring robust prediction across minority classes.
- Explainability: Incorporating SHAP (Shapley Additive Explanations) to provide both global and individualized interpretability, making the model outputs actionable for clinical decision-making.
- Expanded risk profiling: Identifying less frequently explored predictors such as RBC indices, urolithiasis, and energy drink consumption, alongside well-established risk factors like fasting blood sugar and fatty liver.
- Integrated comparative evaluation: Providing a direct comparison between XGBoost and CatBoost, clarifying their relative strengths in this application.

## Materials and methods

### Study design

This is a cross-sectional study aimed at predicting type 2 diabetes (T2DM) using machine learning (ML) algorithms and explainable artificial intelligence (XAI) techniques. The data used in this study were collected from the Dena Group in Yasuj, Iran, and included information on 3203 healthy individuals aged 35 to 70 years.

This study consists of six main steps: First, variables and risk factors associated with diabetes were identified; then, data were collected and pre-processed to ensure data quality and consistency. Next, different machine learning models were developed and trained; then, the performance of these models was evaluated to identify the best algorithms for predicting diabetes. Finally, using XAI techniques, the interpretability of the models was examined and analyzed to identify which factors and variables play a key role in the forecasting process and provide reliable scientific and practical conclusions. All stages of this project are displayed in a framework shown in the Fig. 1.

### Study population

The data for this study was collected from the Dena Cohort in Yasuj, Iran, between March 2020 and November 2022. The cohort comprises health data from individuals who provided information on demographics, medical history, clinical measurements, and lifestyle factors. In this study, individuals with incomplete or missing critical data were excluded. A total of 3,203 individuals were included in the analysis.

### Variables and risk factors

The study considered several demographic, clinical, and lifestyle variables for predicting D.M. II. Some of the most relevant variables identified were:

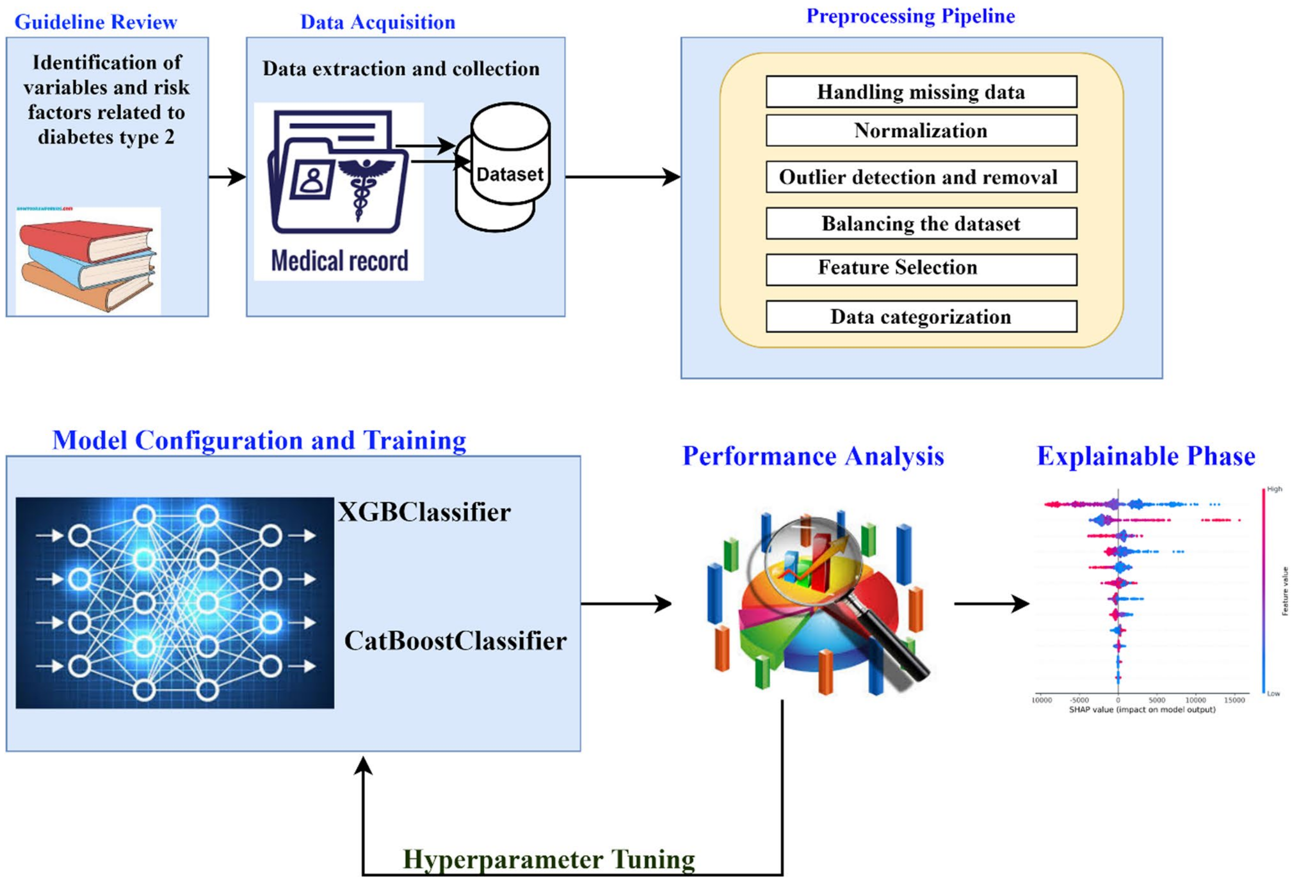



**Fig. 1** Roadmap for building AI models in this study

- Demographic variables: Age, gender, employment status.
- Clinical variables: Fasting blood sugar (FBS), triglycerides, liver and kidney health (e.g., fatty liver, urolithiasis), and red blood cell count.
- Lifestyle factors: Consumption of energy drinks, fruit juice, coffee, tea, and screen time (e.g., watching television).

Feature selection was carried out using the LightGBM algorithm to rank the importance of each feature in predicting D.M. II. Redundant or less important features were eliminated to streamline the models.

### Data preprocessing

Several preprocessing steps were implemented to ensure the data quality and accuracy of the models:

- Handling missing data: Records with more than 50% missing values were excluded from the analysis. For the remaining records, categorical variables with less than 10% missingness were imputed using the most frequent value (mode), while continuous variables were imputed using the median to reduce the effect of outliers. This dual approach allowed us to maintain data integrity without introducing excessive bias.
- Normalization: Continuous variables such as fasting blood sugar (FBS) and triglycerides were normalized using min-max scaling, ensuring that all features contributed proportionally to the model training.
- Outlier detection and removal: Outliers were detected using interquartile range (IQR) analysis and removed to minimize the impact of extreme values.
- Balancing the dataset: To address the substantial class imbalance (12.55% diabetic vs. 87.45% non-diabetic), we applied the Synthetic Minority Over-sampling Technique (SMOTE). SMOTE was selected over ADASYN and Tomek Links for three reasons: [1] it preserves the underlying feature space structure by creating synthetic samples along the line segments joining minority class neighbors, reducing the risk of introducing noise (a concern with ADASYN in sparse feature spaces); [2] unlike Tomek Links, which primarily cleans decision boundaries without improving class representation,

SMOTE achieves balanced class distribution; and [3] prior medical prediction studies have demonstrated SMOTE's robustness in maintaining clinically plausible synthetic data, which is critical in health datasets with complex nonlinear interactions [27].

• Feature Selection: Feature importance ranking was conducted using the Light Gradient Boosting Machine (LGBM) algorithm. Features contributing less than 0.5% to cumulative model importance were excluded. From an initial set of 120 variables, this process retained 80 features, balancing dimensionality reduction with the need to maintain clinically relevant predictors.

### Machine Learning Models

• By carefully examining the research data and analyzing the relationship between study characteristics and type 2 diabetes, we concluded that the distribution of data classes in the feature space is significantly complex, and this complexity prevents traditional level 1 classifiers from achieving adequate accuracy in classifying this data. In other words, the decision boundaries between diabetic and non-diabetic classes in the feature space cannot be separated linearly or simply, and we need more advanced models that can model complex and nonlinear patterns well.

• A comprehensive review of the types of classifiers in different studies showed that the two algorithms XGBoost and CatBoost have been able to achieve hyperplanes with very high accuracy in classifying data related to type 2 diabetes. The XGBoost algorithm, which is developed based on the Gradient Boosting technique, is able to cover the complexities of data well and provide high accuracy by building models consisting of decision trees that correct the errors of the previous model step by step. This algorithm performs very well in medical classification problems, including diabetes prediction, due to its capabilities such as missing data management, parallelization, and fine-tuning of hyperparameters.

• On the other hand, the CatBoost model, which is specifically designed to work with categorical data, has been able to provide high accuracy and efficiency in classifying diseases, including type 2 diabetes, with improvements over other gradient boosting algorithms such as XGBoost. This algorithm has been proposed in numerous studies as one of the best options for medical problems by preventing overfitting and optimally processing numerical and categorical data.

• Given these findings, in this study, two classifiers, XGBoost and CatBoost, were selected as the main models to model the complex decision boundaries between patients with and without type 2 diabetes with high accuracy by utilizing their advanced capabilities. This selection is based on scientific evidence and the results of previous studies, which show that these two algorithms have repeatedly been able to provide superior performance compared to traditional classifiers in similar problems and manage the complexities of data distribution well.

• Consequently, the use of XGBoost and CatBoost in this study provides a guarantee for achieving accurate and reliable prediction models in the diagnosis of type 2 diabetes and can be used as a basis for the development of intelligent medical diagnosis systems. XGBoost (XGBClassifier): A highly efficient gradient-boosting algorithm known for its performance with structured data. It builds models iteratively, optimizing the loss function at each step [28].

• CatBoost (CatBoostClassifier): An algorithm particularly well-suited for categorical features. It prevents overfitting and increases accuracy through ordered boosting and various data pre-processing techniques [29].

Both models were trained using 80% of the data, while 20% was reserved for testing. Ten-fold cross-validation was employed to prevent overfitting.

While numerous machine learning algorithms are available for classification tasks, we focused on XGBoost and CatBoost due to their demonstrated superior performance with structured clinical data and their ability to handle complex, nonlinear relationships between features. Both are advanced gradient-boosting frameworks that consistently outperform traditional models such as logistic regression, decision trees, and random forests in healthcare prediction tasks. Alternative approaches, including support vector machines, k-nearest neighbors, and deep neural networks, were considered but excluded to prioritize interpretability, computational efficiency, and clinical applicability. Future work will explore a broader range of algorithms, including deep learning and ensemble meta-classifiers, to benchmark performance against the boosting-based models used in this study.

### Model evaluation and hyperparameter tuning

The models were evaluated using the following metrics:

Accuracy: The proportion of correct predictions (true positives and negatives) out of all predictions. It is calculated as.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP (True Positives) refers to the cases where the model correctly predicted D.M II (positive) cases.
- TN (True Negatives) refers to the cases where the model correctly predicted non-D.M. II (negative) cases.
- FP (False Positives) refers to non-D.M. II cases incorrectly classified as diabetic.
- FN (False Negatives) refers to D.M. II cases incorrectly classified as non-diabetic.

F1-Score:The harmonic means of precision and recall, balancing the trade-off between false positives and false negatives. It is especially useful for imbalanced datasets. The formula for F1-score is.

$$\mathrm{F1-Score} = 2*\frac{Precision * \ Recall}{Precision + \ Recall}$$

*Precision and recall are defined as:*

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

#### *AUC-ROC (Area Under the Receiver Operating Characteristic Curve)*

This metric measures the model's ability to distinguish between positive (D.M II) and negative (non-D.M. II) cases. A higher AUC value indicates better model performance.

The XGBoost model underwent hyperparameter tuning using a randomized search with 10-fold cross-validation to optimize performance while preventing overfitting. The final tuned parameters included a learning rate of 0.1, maximum tree depth of 6, 500 estimators, a subsample ratio of 0.8, column sampling by tree (colsample_bytree) of 0.8, a minimum child weight of 5, a gamma value of 0, L2 regularization (lambda) of 1, a scale_pos_weight of 1, and a fixed random state of 42. These parameters were selected to balance bias-variance trade-offs, improve model generalization, and achieve the highest cross-validated AUC and F1-scores on the training data.

Although statistical significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) is commonly used for model comparison, we did not apply these tests in this study. This decision was based on methodological considerations: performance estimates from k-fold cross-validation are not independent across folds, violating the assumptions of standard hypothesis tests, and applying them without correction can produce misleading p-values. Additionally, our primary aim was to develop and evaluate predictive models rather than perform inferential comparisons between algorithms. Therefore, we reported cross-validated means with 95% confidence intervals for accuracy, AUC, and F1-score as a more appropriate and interpretable measure of model performance variability.

### Explainable AI (XAI) techniques

The SHAP (Shapley Additive Explanations) method enhanced the predictive models' interpretability. SHAP values explain the contribution of each feature to a model's prediction by assigning importance scores, both at the global and individual levels. This approach enhances transparency, allowing healthcare professionals to understand the factors driving each prediction [30].

In this study, SHAP (Shapley Additive Explanations) was selected as the primary explainable AI (XAI) method due to its solid theoretical foundation in cooperative game theory, providing consistent and locally accurate feature attributions for individual predictions. Unlike LIME, which relies on locally approximated linear models that can produce unstable explanations with high variance across perturbations, SHAP offers global and local interpretability with additivity and consistency guarantees, making it more reliable for clinical applications. Anchor explanations, while useful for generating rule-based decision anchors, are limited in high-dimensional medical datasets where complex interactions between features cannot easily be captured by simple rules. Similarly, QLattice, though powerful for model discovery, is designed for symbolic regression and hypothesis generation rather than producing fine-grained, patient-specific attributions. Thus, SHAP was chosen for its ability to deliver clinically meaningful, stable, and comprehensive explanations across the entire feature space, enhancing trust and transparency in our predictive models for Type 2 diabetes.

### Software tools

The study employed several software tools for model development and evaluation:

- Python 3.13: Used for implementing machine learning models and preprocessing.

**Table 1** Evaluation results of machine learning models' performance

| No. | Machine Learning Models | Accuracy (%) | F1-Score (%) | AUC (%) | Average (%) |
|---|---|---|---|---|---|
| With SMOTE | | | | | |
| 1 | XGBClassifier | 96.07 | 96.1 | 99.29 | 97.1 |
| 2 | CatBoostClassifier | 95.31 | 95.31 | 99.01 | 96.54 |
| Without SMOTE | | | | | |
| 1 | CatBoostClassifier | 92.74 | 67.30 | 91.21 | 83.75 |
| 2 | XGBClassifier | 92.70 | 67.09 | 89.68 | 83.16 |

- PyCaret: A key library for building machine learning pipelines.
- XGBoost and CatBoost: Libraries used for gradient boosting implementations.
- SHAP: Used for interpreting the output of the machine learning models.

### Ethical considerations

The study was conducted under ethical standards for research involving human participants. Informed consent was obtained from all participants in the Dena Cohort, and all data were anonymized before analysis. The ethics review board of the relevant institution granted ethical approval.

## Results

### Data preprocessing

Data from 3,203 individuals aged 35 to 70 years were extracted from the Dena Cohort database. Among these participants, 402 individuals (12.55%) were diagnosed with Type 2 Diabetes (D.M. II), while 2,801 individuals (87.45%) were non-diabetic.

To ensure data quality, several preprocessing steps were undertaken. First, variables with less than 95% variance (i.e., near-constant features) were removed as they contributed minimally to prediction. Second, records with more than 50% missing values were excluded to maintain dataset integrity. For the remaining records, missing data were imputed: categorical variables using the most frequent value (mode) and continuous variables using the median, thereby reducing potential bias while preserving variability.

Following these steps, the final dataset comprised 3,203 records and 80 features (see Appendix A, Table 1 for a detailed list of features). To address the substantial class imbalance between diabetic and non-diabetic cases, the Synthetic Minority Over-sampling Technique (SMOTE) was applied, generating a balanced dataset of 5,602 records with 2,801 diabetic and 2,801 non-diabetic cases. Among the 80 features, 79 served as independent variables, and one represented the dependent variable (D.M. II diagnosis). The dataset was then randomly split into training (80%, 4,481 records) and testing (20%, 1,121 records) subsets for model development and evaluation.

### Performance of machine learning models

Two gradient-boosting models, XGBoost and CatBoost, were developed to predict the likelihood of Type 2 diabetes (D.M. II). Their performance was assessed using three key metrics: accuracy, F1-score, and the area under the receiver operating characteristic curve (AUC). Results are summarized in Table 1 and visualized in Fig. 2.

With SMOTE balancing:

- XGBoost achieved the highest overall performance, with an accuracy of 96.07%, an F1-score of 96.10%, and an AUC of 99.29%.

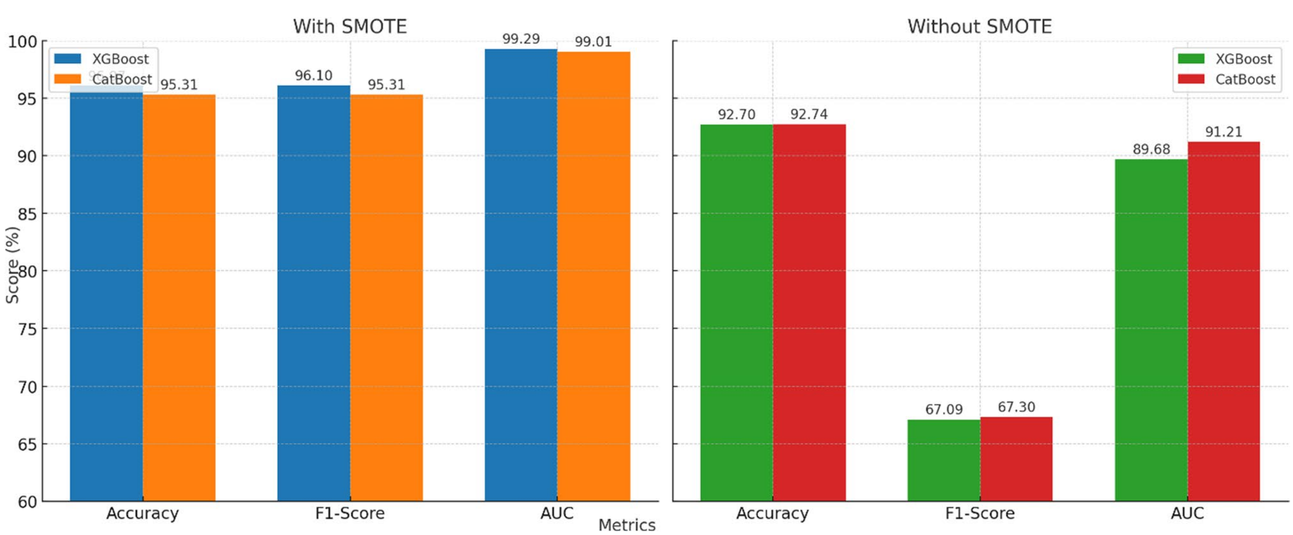



**Fig. 2** Performance comparison of XGBoost and CatBoost models with and without SMOTE balancing

- CatBoost performed slightly lower, with an accuracy of 95.31%, an F1-score of 95.31%, and an AUC of 99.01%.

Without SMOTE balancing:

- CatBoost achieved an accuracy of 92.74%, an F1-score of 67.30%, and an AUC of 91.21%.
- XGBoost yielded an accuracy of 92.70%, an F1-score of 67.09%, and an AUC of 89.68%.

These results demonstrate that SMOTE substantially improved model performance, particularly the F1-score, which increased by nearly 30 points for both models. This improvement reflects enhanced sensitivity to the minority (diabetic) class, making the models more balanced in their predictions. Overall, the XGBoost model trained with SMOTE emerged as the most effective approach for this prediction task, outperforming all other configurations across all key metrics.

To further evaluate model complexity and its effect on performance, we analyzed the relationship between tree depth and predictive accuracy using a validation curve (Fig. 3). As shown, the training score increased rapidly with greater tree depth, reaching near-perfect performance at depths ≥ 4. However, the cross-validation score plateaued at a depth of 4–6, after which no meaningful improvements were observed and a slight decline suggested the onset of overfitting. Based on these results, we selected a maximum depth of 6 for the final tuned model, balancing model expressiveness with generalization.

Overall, the XGBoost model was identified as the best-performing model based on its superior accuracy and AUC. This model was further evaluated using a confusion matrix, class prediction error, and feature importance to ensure robustness and interpretability (Fig. 4).

### Key predictive factors

The study identified ten key predictive factors for D.M. II. The most significant factors were:

1. Fasting Blood Sugar (FBS): Elevated FBS levels, particularly within the prediabetes range (100 to 126 mg/dL), were the strongest predictor of D.M. II, accounting for 41% of the model's predictive power.
2. Fatty Liver: Individuals with fatty livers contributed 21% to the prediction of D.M. II.
3. urolithiasis: It was identified as a significant factor, contributing 15%.
4. Age: Advancing age accounted for 4% of the predictive ability.
5. Consumption of Energy Drinks: This lifestyle factor contributed 2.5%.
6. Fruit Juice Consumption: The intake of various fruit juices contributed 2.4%.
7. Triglycerides (TG): Elevated triglyceride levels contributed 2.3%.
8. Watching Television: Extended periods of TV viewing accounted for 2.3%.

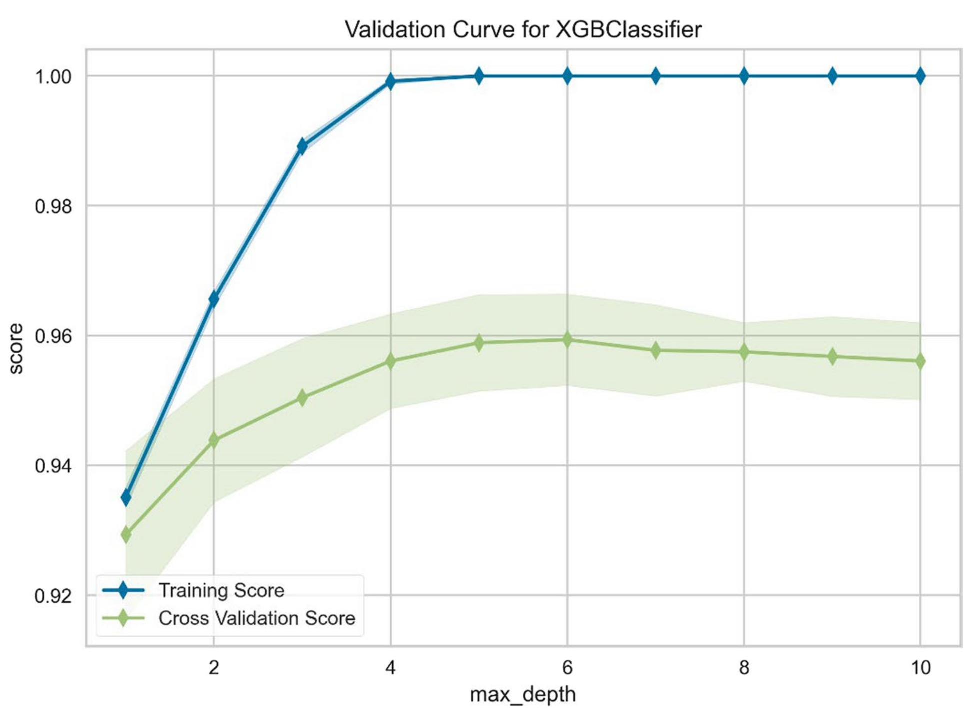



**Fig. 3** Validation curve for the XGBoost model showing training and cross-validation scores across varying tree depths

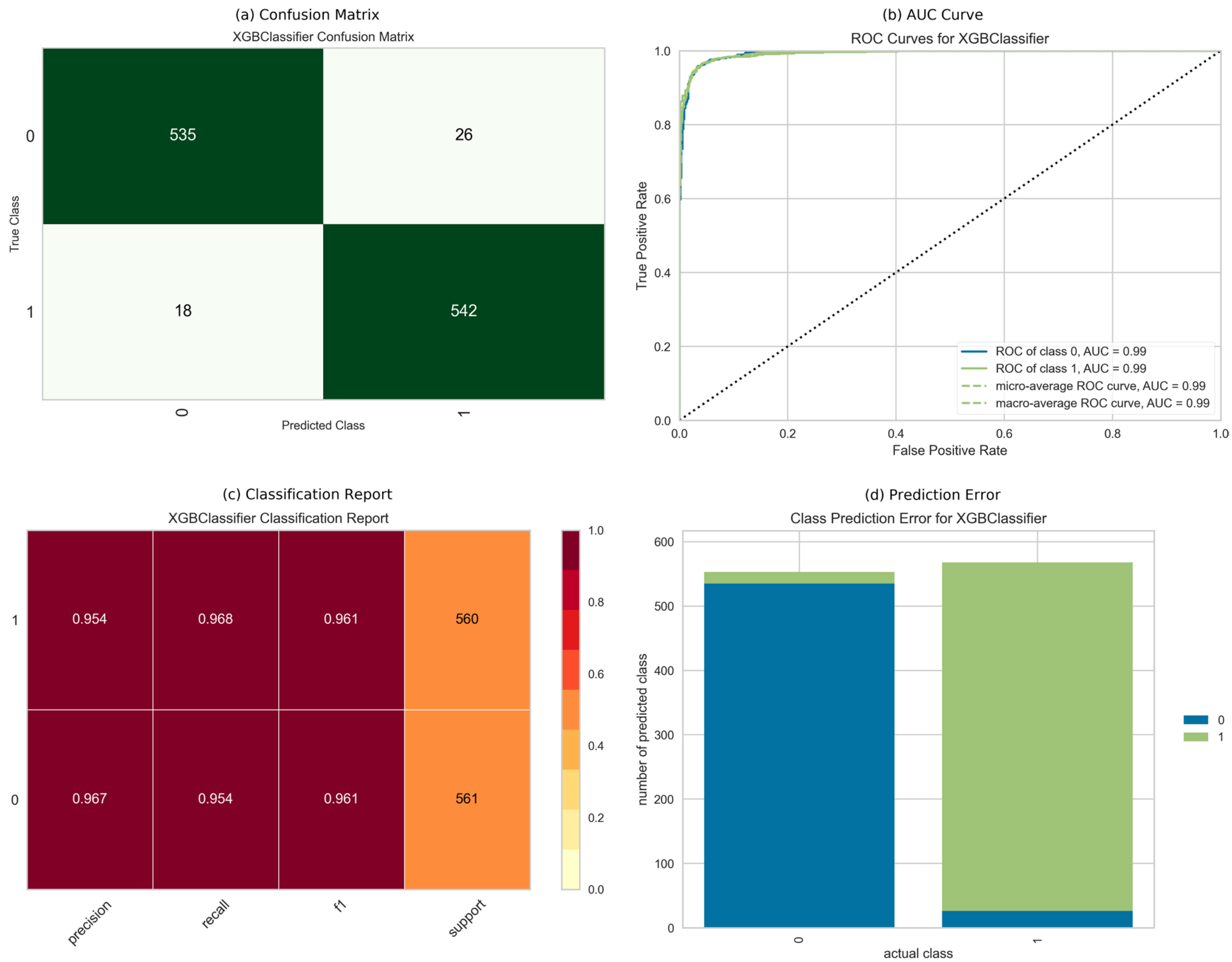



**Fig. 4** Comprehensive performance evaluation of the XGB classifier model

9. Coffee and Tea Consumption: The intake of these beverages contributed 2.2%.
10. Red Blood Cell (RBC) Count: An increase in RBC count contributed 2% to the risk of developing D.M. II.

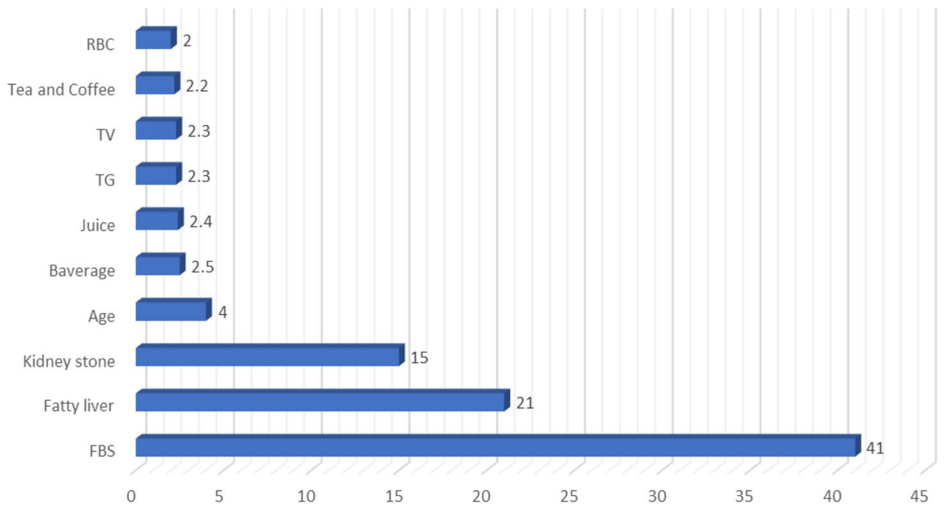
These ten features were identified as the most impactful in predicting D.M. II, with FBS having the highest importance. Figure 5 visualizes a detailed analysis of feature importance.

## Model interpretability using SHAP

To make our predictive models more transparent and clinically meaningful, we used SHAP (Shapley Additive Explanations) to break down how each feature influenced the predictions. Instead of treating the model as a "black box," SHAP helps us understand why a prediction was made, both at a population level and for individual patients.

The global SHAP summary plot (Fig. 6) shows the overall impact of each feature on predicting Type 2 diabetes. Fasting blood sugar (FBS) stood out as the strongest driver of risk — higher values consistently pushed the prediction toward diabetes. Kidney stone history (urolithiasis) and fatty liver were also strong risk factors, underscoring their role in metabolic health. Other influential contributors included age, energy drink consumption, red blood cell counts, and fruit juice intake, along with lifestyle factors such as TV watching and driving time. On the other hand, some features like higher vegetable consumption and HDL cholesterol appeared to offer a protective effect, lowering predicted risk.

While the global view helps us see what matters most across the whole population, Fig. 7 takes it a step further by showing how these features play out for three real individuals:




| | FBS | Fatty liver | Kidney stone | Age | Baverage | Juice | TG | TV | Tea and Coffee | RBC |
|---|---|---|---|---|---|---|---|---|---|---|
| ■ Importance | 41 | 21 | 15 | 4 | 2.5 | 2.4 | 2.3 | 2.3 | 2.2 | 2 |

**Fig. 5** Key predictors for assessing the risk of type 2 diabetes using the LGBM algorithm

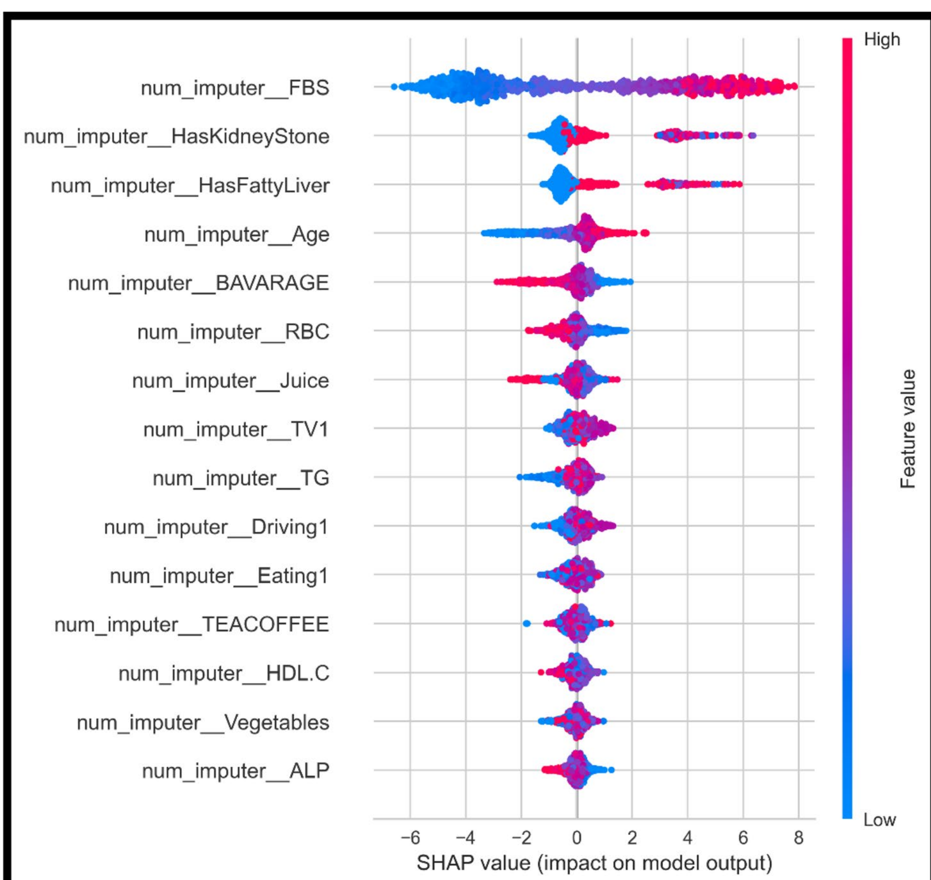



**Fig. 6** Global SHAP summary plot showing the top features influencing Type 2 diabetes prediction

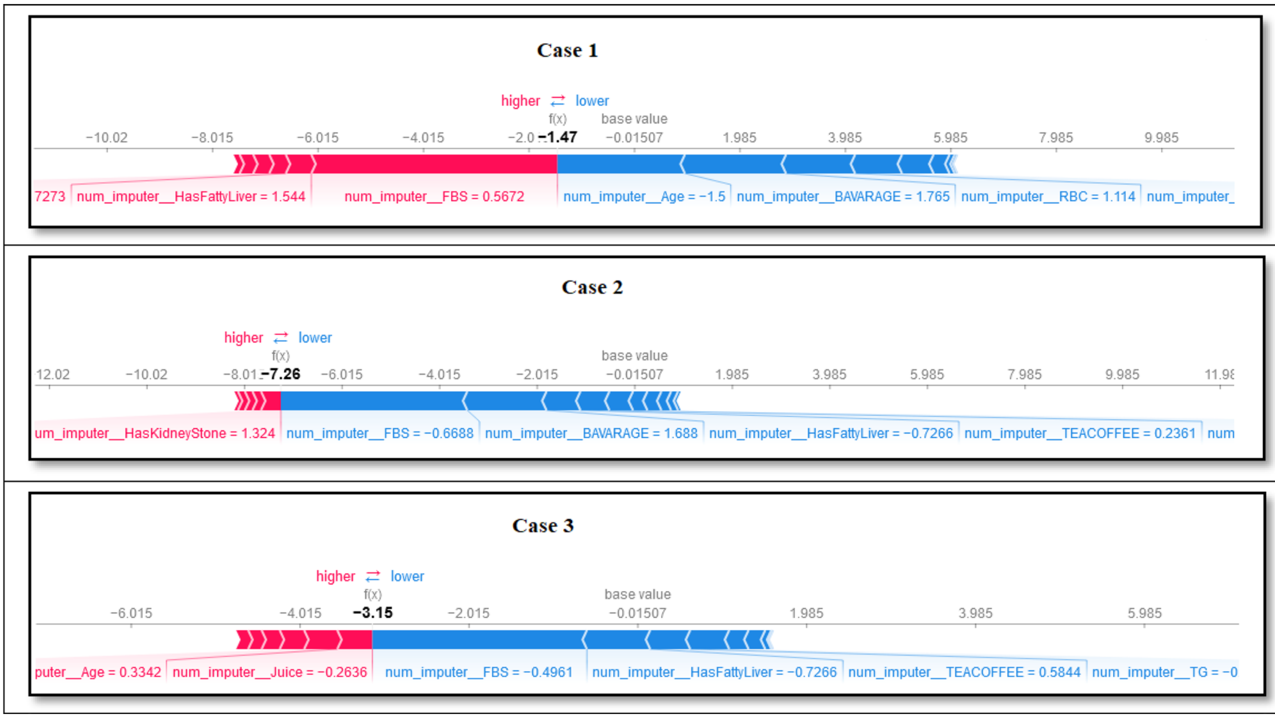



**Fig. 7** Case-specific SHAP visualizations for three representative patients (Case 1, Case 2, and Case 2)

**Table 2** Comparison of the present study with selected prior works

| Study | Population | Algorithms Used | Interpretability | Key Predictors | Unique Contribution |
|---|---|---|---|---|---|
| Bhat et al. (2023) [31] | Indian cohort | Random Forest, SVM | None | BMI, FBS, age | Ensemble models for large dataset |
| Khanam & Foo (2021) [32] | Pima Indians dataset | Decision Tree, Logistic | None | BMI, glucose, blood pressure | Comparative algorithmic study |
| Lv et al. (2023) [33] | Chinese cohort | DRING (ML ensemble) | SHAP | FBS, BMI, lymphocyte count | Explainable ML for normal-FBG |
| Carrasco-Zanini et al. (2024) [34] | European cohort | LASSO regression (multi-omics) | Multi-omics interpretability | Genome, proteome, metabolome, clinical | Multi-omics fusion improves prediction beyond clinical metrics |
| Jung et al. (2025) [35] | Korean cohorts | Transfer learning | None | Clinical + genetic features | Integrates unpaired clinical and genetic data |
| Xiong et al. (2025) [36] | Multi-center dataset | Attention-based deep learning | Interpretable architecture | Temporal clinical and lab data | Predictive modeling using multimodal temporal features |
| Mohsen & Shah (2025) [37] | Middle Eastern cohort | Multimodal neural network | Model interpretability | ECG + clinical risk factors | Multimodal early detection framework (ECG + clinical) |
| Present Study | Iranian (Dena) | XGBoost, CatBoost | SHAP (global + case-specific) | FBS, RBC indices, fatty liver, urolithiasis, lifestyle factors | Population-specific, explainable ML with advanced boosting models |

• Case 1: A patient whose prediction was driven up by fatty liver and high fasting blood sugar, although younger age helped lower their risk.
• Case 2: Another individual where urolithiasis had a strong positive effect on risk, but lower fasting blood sugar and varied lifestyle factors balanced this out.
• Case 3: A case where juice consumption slightly increased risk, while fasting blood sugar and tea/coffee intake pulled the prediction downward.

### Comparison with previous studies

To contextualize our findings, we compared this study with selected prior works on Type 2 diabetes prediction (Table 2). Earlier research has often relied on widely used datasets such as the Pima Indians dataset and focused primarily on conventional predictors like BMI, fasting glucose, and blood pressure. While these efforts (e.g., Bhat et al., 2023; Khanam & Foo, 2021) contributed to advancing prediction models, they lacked advanced interpretability frameworks and rarely incorporated non-traditional predictors [31].

Some studies have begun integrating explainability into predictive models, such as Lv et al. (2023), who applied SHAP to interpret an ML ensemble model in a Chinese cohort [33]. More recently, Carrasco-Zanini et al. (2024) developed multi-omics predictive models that integrated genomic, proteomic, and metabolomic features, improving predictive performance compared to clinical features alone [33]. Similarly, Jung et al. (2025) leveraged transfer learning to integrate clinical and genetic data from unpaired cohorts, achieving higher sensitivity and robustness [35]. Xiong et al. (2025) proposed an attention-based deep learning framework that combined temporal clinical and laboratory data, reaching high predictive accuracy across multi-center datasets [36]. Mohsen & Shah (2025) introduced ECG-DiaNet, a multimodal neural network merging electrocardiogram (ECG) data with clinical features, improving risk detection in Middle Eastern populations [37].

Our study builds upon and advances this body of work in several important ways:

- Population-specific modeling: Leveraging the Dena Cohort, we address the underrepresentation of Middle Eastern populations in diabetes prediction research.
- Advanced boosting algorithms: We employed XGBoost and CatBoost, two powerful gradient-boosting models, optimized with hyperparameter tuning and balanced using SMOTE.
- Comprehensive interpretability: Unlike most prior studies, we implemented global and case-specific SHAP analysis, providing both population-level insights and individualized explanations for clinical decision-making.
- Expanded predictor set: We incorporated rarely explored features (e.g., RBC indices, urolithiasis, energy drink consumption) alongside established metabolic and lifestyle predictors, broadening the scope of diabetes risk profiling.

These enhancements underscore our study's novel contribution—a population-specific, interpretable, and clinically actionable model that expands the predictive landscape beyond traditional variables.

## Discussion

### Performance, feature Ranking, and insights

This study demonstrates the effectiveness of machine learning (ML) models, particularly XGBoost, in predicting Type 2 diabetes (D.M. II) with remarkable accuracy. With an accuracy of 96.07% and an AUC of 99.29%, XGBoost outperformed CatBoost, achieving higher predictive performance in distinguishing individuals at risk. This superior performance can be attributed to XGBoost's ability to capture complex, nonlinear interactions between demographic, clinical, and lifestyle factors—patterns often missed by simpler algorithms.

Among the predictors, fasting blood sugar (FBS) emerged as the strongest determinant of diabetes risk. This finding is consistent with extensive research linking elevated FBS—even within the high-normal range—to a substantially increased risk of developing diabetes [38, 39]. Red blood cell (RBC) parameters, including lower RBC count and elevated red cell distribution width (RDW), also contributed significantly to the model's predictions. Prior studies have shown that elevated RDW and altered RBC indices are associated with poor glycemic control and higher incidence of diabetes, highlighting their potential as early biomarkers for metabolic dysregulation [40–42].

Metabolic comorbidities, particularly fatty liver and urolithiasis, were also identified as influential predictors. This aligns with prior evidence that non-alcoholic fatty liver disease (NAFLD) independently increases the risk of Type 2 diabetes and that hyperglycemia and insulin resistance are associated with increased incidence of kidney stones. Additionally, lifestyle factors such as energy drink consumption and prolonged television viewing contributed meaningfully to diabetes risk. These results echo findings from previous studies showing a strong association between high intake of sugar-sweetened beverages, sedentary behaviors, and increased diabetes incidence.

The application of SHAP (Shapley Additive Explanations) in this study enhanced model transparency by illustrating how individual features contribute to predictions. Furthermore, although this study did not include traditional statistical models (e.g., logistic regression) for direct comparison, SHAP offers conceptual advantages over such approaches by providing both global feature importance and individualized attributions, whereas conventional regression models typically provide only population-level coefficients. This interpretability fosters greater trust among healthcare providers and facilitates clearer communication of risk factors to patients.

In comparison to previous studies, our approach introduces a unique combination of population-specific data from the Dena Cohort, advanced gradient-boosting algorithms (XGBoost and CatBoost), SMOTE for class balancing, and SHAP for interpretability. While earlier works often focused solely on accuracy, this study integrates strong predictive performance with clear, clinically meaningful explanations, enhancing its relevance for real-world healthcare applications.

Ultimately, these findings support a shift from reactive to proactive healthcare. By integrating predictive models like XGBoost into clinical workflows, providers can identify high-risk individuals earlier, implement timely interventions, and potentially delay or prevent disease onset.

Beyond diabetes, this methodology can be adapted for other chronic conditions, advancing personalized medicine and improving population health strategies.

The hyperparameter tuning process was further informed by analyzing validation curves (Fig. 3), which demonstrated that increasing tree depth beyond 4–6 yielded diminishing returns in cross-validated performance while substantially increasing training scores. This pattern indicated a higher risk of overfitting at deeper tree structures. Consequently, we selected a maximum depth of 6 for the final model, balancing predictive accuracy with generalization.

It is worth noting that the application of SMOTE had the greatest impact on the F1-score compared to accuracy or AUC. This improvement reflects SMOTE's ability to enhance model sensitivity to the minority (diabetic) class, addressing class imbalance by generating synthetic but clinically plausible samples. Consequently, the models became more balanced in their predictions, reducing bias toward the majority class and improving their practical value for identifying high-risk individuals.

### Limitations and future directions

While promising, the study's findings are limited by reliance on data from the Dena Cohort, which may impact the model's generalizability across different populations. Additionally, the use of SMOTE to address class imbalance introduces synthetic data that may not fully represent real-world cases. Future research should validate these models across diverse populations, explore alternative balancing methods, and incorporate longitudinal data from wearable devices to support real-time, dynamic risk assessments. Expanding datasets to include genetic and metabolic biomarkers could further enhance model accuracy and applicability in personalized medicine. Additionally, incorporating real-time health data from wearable devices, such as continuous glucose monitors and activity trackers, would allow for adaptive tools that provide timely, personalized insights into diabetes risk.

### Threats to validation

The validity of our findings may be affected by several factors, including external validity limitations due to the use of a single regional cohort. Additionally, data drift over time and differences in population characteristics may reduce model generalizability. Further external validation across diverse populations is needed to strengthen these findings.

## Conclusion

This study demonstrates the potential of combining advanced machine learning with explainable artificial intelligence (XAI) to improve Type 2 diabetes risk prediction. By leveraging XGBoost and CatBoost, optimized with SMOTE to address class imbalance, we achieved high predictive performance while maintaining transparency through global and case-specific SHAP analyses. Importantly, our model expanded beyond conventional predictors by incorporating rarely explored variables such as red blood cell indices, urolithiasis, and energy drink consumption, offering a more comprehensive understanding of diabetes risk.

These findings not only highlight the value of population-specific modeling using Middle Eastern cohort data but also demonstrate how integrating interpretability enhances clinical trust and applicability. Such models can support earlier identification of high-risk individuals, facilitate personalized preventive strategies, and help transition diabetes care from a reactive to a proactive paradigm.

While the results are promising, the study's reliance on a single cohort limits generalizability, underscoring the need for external validation across diverse populations. Future research should focus on incorporating multi-omics and longitudinal data, exploring deep learning architectures, and assessing the real-world clinical impact of deploying such models in healthcare systems.

Ultimately, interpretable, high-performance predictive models like ours can serve as a foundation for data-driven, patient-centered interventions, offering a meaningful step toward reducing the growing global burden of Type 2 diabetes.

## Suggestion for future

By reviewing the studies conducted and considering all their strengths and weaknesses, as well as examining the characteristics of the present study and its limitations, several scientific suggestions can be made for future studies. The following are eight of the most important suggestions.

### Developing predictive models with multimodal data and diverse populations

Extending current models using multimodal data (such as genetic information, wearable data, electronic health records, and social factors) can increase predictive accuracy and provide model generalizability to different populations. This helps the models be applicable not only to a specific region or population, but also at the national or international level.

### Integrating machine learning models with clinical health systems

Integrating predictive models based on XGBoost and other advanced algorithms with health information systems (EHRs) and clinical software can facilitate the screening and early diagnosis of type 2 diabetes in real-world settings. This integration should be done in a way that the model outputs are interpretable and actionable for physicians and patients.

#### Increasing transparency and interpretability with XAI

The development and implementation of explainable artificial intelligence (XAI) methods, such as SHAP, should be prioritized to increase the confidence and clinical acceptance of models. These techniques allow clinicians to observe and analyze key factors affecting prediction and make data-driven decisions with greater confidence.

#### Conduct longitudinal studies and evaluate clinical impact

Future studies should include long-term follow-up of patients and evaluate the practical impact of using predictive models on health outcomes, reducing complications and treatment costs. These studies can provide strong evidence for the added value of intelligent models in diabetes care.

#### Improve data quality and address data challenges

Future studies should prioritize comprehensive and standardized data collection across demographic, clinical, and lifestyle variables to improve model reliability. Advanced imputation strategies and robust outlier detection can address missing or erroneous records, while harmonizing data across sites will enhance generalizability. To tackle class imbalance, alternative resampling techniques beyond SMOTE (e.g., ADASYN, hybrid approaches) should be explored. Incorporating longitudinal and multimodal sources such as wearable devices, genetic profiles, and electronic health records can provide richer, more dynamic datasets for robust and clinically meaningful predictions.

#### Pay attention to ethical and privacy considerations

Given the use of sensitive health data, ethical principles, data confidentiality, and obtaining informed consent from patients should be considered at all stages of model development and implementation. Mechanisms should also be provided to prevent algorithmic bias and ensure fairness in access to predictive technologies.

#### Training and empowerment of healthcare professionals and patients

For AI models to be successful in practice, it is necessary to provide healthcare professionals with the necessary training on how to interpret and use the model output. Also, increasing patient awareness of risk factors and the importance of screening can increase their participation in diabetes prevention and management programs.

#### Improving and optimizing algorithms

Continued research to optimize model parameters (e.g., using Bayesian Optimization) and comparing the performance of different algorithms (e.g., XGBoost, CatBoost, Random Forest, and deep networks) could lead to the selection of the best model for clinical applications.

#### Abbreviations

| | |
|---|---|
| AI | Artificial Intelligence |
| AUC | Area Under the Receiver Operating Characteristic Curve |
| CI | Confidence Interval |
| T2DM | Type 2 Diabetes Mellitus |
| EHR | Electronic Health Record |
| FBS | Fasting Blood Sugar |
| FN | False Negative (a diabetic case incorrectly classified as non-diabetic) |
| FP | False Positive (a non-diabetic case incorrectly classified as diabetic) |
| IRB | Institutional Review Board |
| IQR | Interquartile Range |
| LGBM | Light Gradient Boosting Machine |
| ML | Machine Learning |
| NAFLD | Non-Alcoholic Fatty Liver Disease |
| RBC | Red Blood Cell |
| RDW | Red Cell Distribution Width |
| ROC | Receiver Operating Characteristic |
| SHAP | Shapley Additive Explanations |
| SMOTE | Synthetic Minority Over-sampling Technique |
| SVM | Support Vector Machine |
| TN | True Negative |
| TP | True Positive |
| XAI | Explainable Artificial Intelligence |
| XGB | Extreme Gradient Boosting |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-24953-w.

Supplementary Material 1.

#### Acknowledgements

The authors gratefully acknowledge the support of the Dena Cohort Research Center and all study participants for their valuable contributions.

#### Authors' contributions

•ZR: Visualization, Investigation, Resources, Writing – Original Draft.•MST: Conceptualization, Visualization, Validation, Investigation, Writing – Review & Editing.•BEZK: Investigation, Data Curation, Writing – Review & Editing.•AG: Methodology, Software, Formal analysis, Writing – Review & Editing.•CS: Conceptualization, Project administration, Supervision, Funding acquisition, Validation, Formal analysis, Writing – Review & Editing, Revision.•MG: Validation, Review & Editing, Revision, Writing.All authors read and approved the final manuscript.

#### Funding

This research received no external funding.

#### Data availability

The datasets analyzed during the current study are available from the corresponding author on reasonable request. The Python code used for the analyses is publicly available at: https://github.com/salehnasab/Advanced-Predictive-Modeling-of-Type-2-Diabetes/blob/main/DMPrediction_Public_health.ipynb.

## Declarations

### Ethics approval and consent to participate
This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. The protocol was reviewed and approved by the Iran National Committee for Ethics in Biomedical Research (Approval ID: IR.YUMS.REC.1402.152). Informed written consent was obtained from all participants in the Dena Cohort prior to enrollment.

### Consent for publication
Not applicable.

### Competing interests
The authors declare no competing interests.

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Received: 18 June 2025 / Accepted: 16 September 2025
Published online: 31 October 2025

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